Sentinel AK-XL - Virtual SOC

Project Overview & Technical Specification

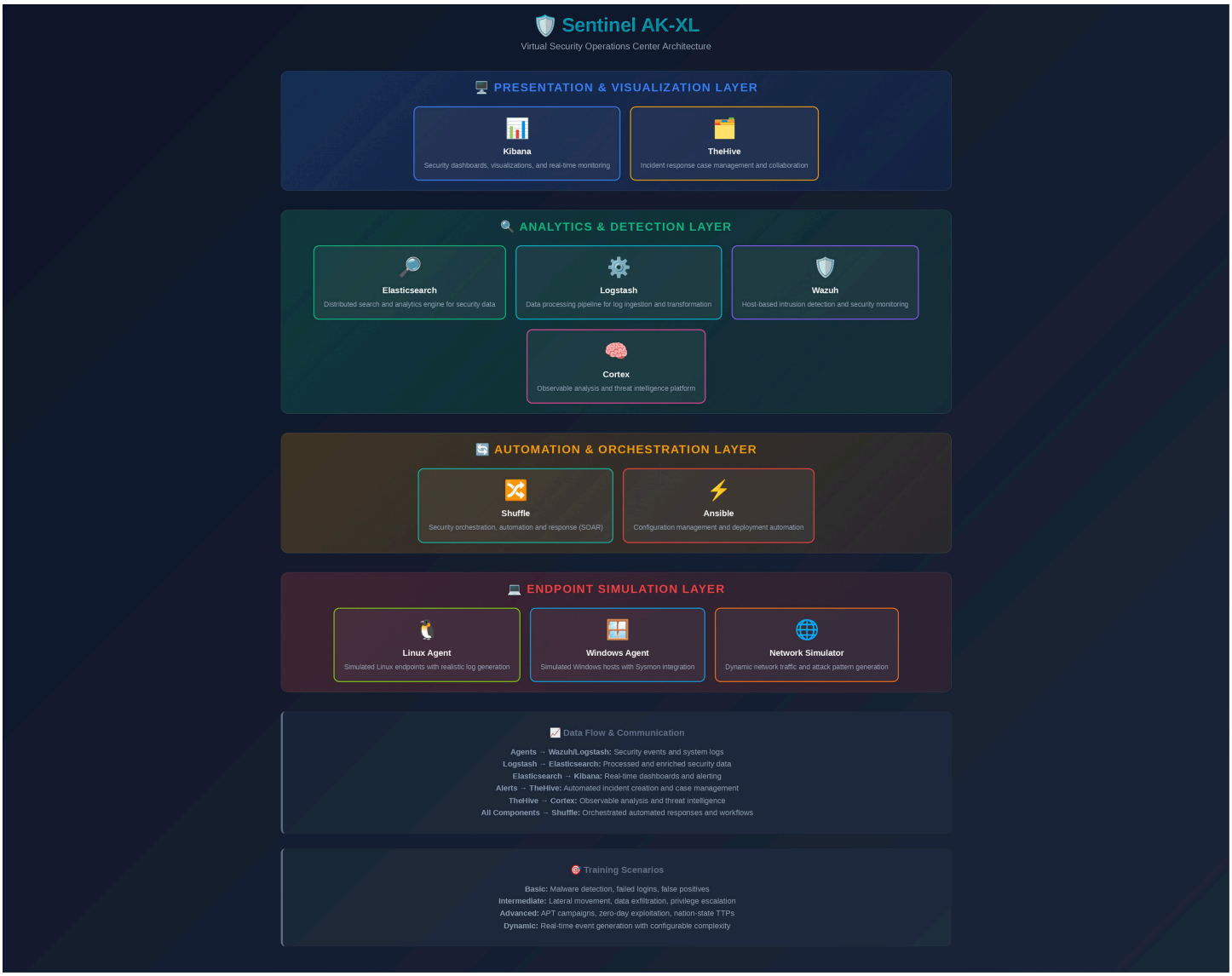
Executive Summary

Sentinel AK-XL is a comprehensive containerized Virtual Security Operations Center (SOC) designed to provide hands-on training for blue team analysts in threat detection and incident response. The platform simulates real-world enterprise environments with dynamic threat scenarios, enabling security professionals to develop skills in a controlled, reproducible environment.

Project Objectives

- **Training Environment**: Create realistic SOC scenarios for analyst skill development
- **Incident Response**: Provide comprehensive incident handling workflows and playbooks
- **Scalable Deployment**: Enable easy setup across different environments with minimal configuration
- **Dynamic Threat Simulation**: Generate real-time security events with varying complexity levels

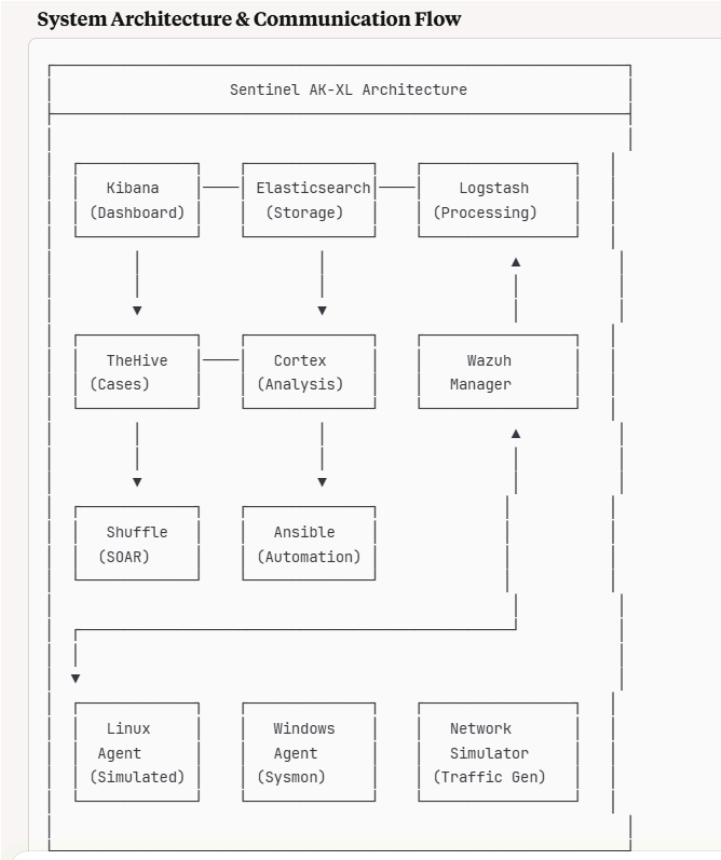
Technology Stack & Component Functions



Core SOC Infrastructure

- **ELK Stack (Elasticsearch, Logstash, Kibana)**
  - *Function*: Centralized log collection, processing, and visualization
  - *Role*: Primary SIEM functionality, dashboard creation, and data analytics
- **Wazuh**
  - *Function*: Host-based intrusion detection system (HIDS) and security monitoring
  - *Role*: Endpoint monitoring, file integrity monitoring, vulnerability detection
- **TheHive**
  - *Function*: Security incident response platform and case management
  - *Role*: Incident tracking, collaboration, and workflow management
- **Cortex**
  - *Function*: Observable analysis and threat intelligence platform
  - *Role*: Automated artifact analysis, threat enrichment, and IOC processing
- **Shuffle**
  - *Function*: Security Orchestration, Automation and Response (SOAR)
  - *Role*: Automated incident response workflows and playbook execution
- **Ansible**
  - *Function*: Configuration management and automation

- *Role*: Environment provisioning and scenario deployment
- **Sysmon**
  - *Function*: Enhanced Windows logging for security monitoring
  - *Role*: Detailed process and network activity logging on simulated Windows endpoints



Data Flow & Scenarios

Training Scenarios

1. **Basic Level**:
   - Malware detection alerts
   - Failed authentication attempts
   - Basic network anomalies
   - False positives from legitimate admin activities
2. **Intermediate Level**:
   - Lateral movement patterns
   - Data exfiltration attempts
   - Privilege escalation events
   - Multi-stage attack simulations
3. **Advanced Level**:
   - Advanced Persistent Threat (APT) campaigns
   - Zero-day exploitation scenarios
   - Supply chain attack simulations
   - Nation-state actor TTPs

Data & Alert Flow

Sentinel AK-XL Virtual SOC - Real-time Security Event Processing

1

Simulated Endpoints
Windows & Linux hosts generate security events

System Logs | Sysmon Events | Network Traffic | Process Activity

Raw Events
Continuous stream of security telemetry

2

Wazuh Agents
Collect & forward endpoint data

Data Processing
- Log normalization
- Event correlation
- Threat detection rules
- Data enrichment

Logstash
Parse, transform & enrich logs

3

Elasticsearch
Distributed search & analytics engine

Indexing & Storage
- Full-text search indexing
- Time-series data optimization
- Data retention policies
- Real-time aggregations

Kibana
Visualization & dashboard platform

4

Detection Rules
Wazuh rules + Custom detection logic

CRITICAL
Malware, Data Breach

HIGH
Lateral Movement

MEDIUM
Suspicious Activity

LOW
Policy Violations

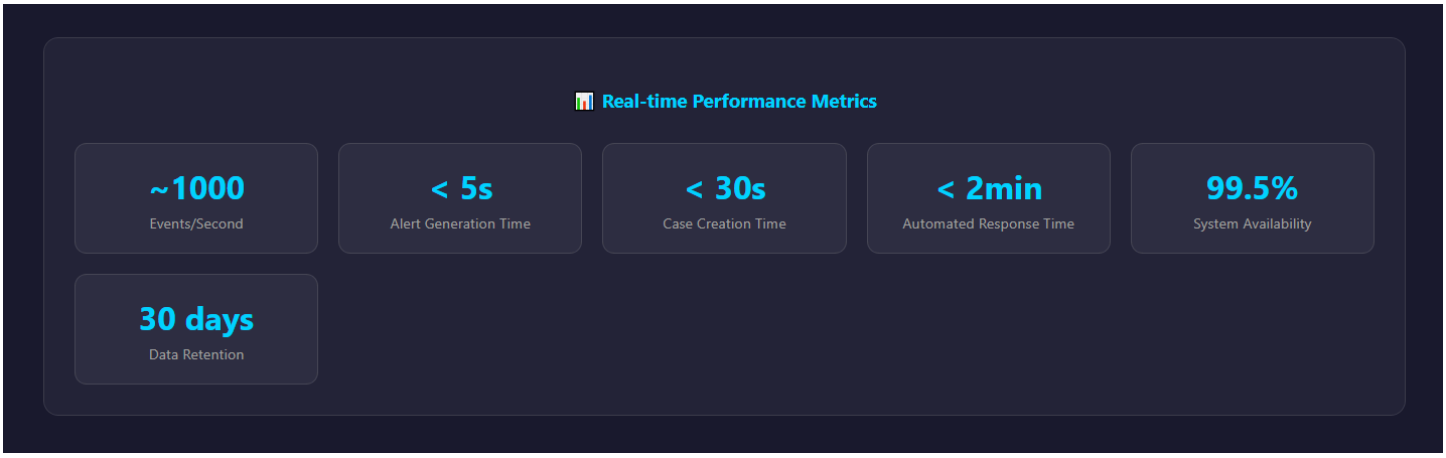
Alert Queue
Prioritized security alerts

5

TheHive
Automated case creation from alerts

Case Management
- Alert deduplication
- Case assignment
- SLA tracking
- Evidence collection

Cortex
Observable analysis & threat intel



Dynamic Data Generation

- **Real-time Log Streams**: Continuous generation of realistic security events
- **Threat Intelligence**: Updated IOCs and threat signatures
- **Network Traffic**: Simulated normal and malicious network patterns
- **Endpoint Behavior**: Realistic user and system activity patterns

Project Repository Structure

```
sentinel-ak-xl/
├── README.md                    # Project documentation and setup guide
├── docker-compose.yml           # Main container orchestration
├── setup.sh                     # Initial environment setup script
├── run-scenario.sh              # Scenario management script
├── cleanup.sh                   # Environment reset script
├── docs/                        # Documentation directory
│   ├── user-guide.md           # Analyst training guide
│   ├── admin-guide.md          # Administrative documentation
│   └── troubleshooting.md      # Common issues and solutions
├── configs/                     # Configuration files
│   ├── elk/                    # ELK stack configurations
│   ├── wazuh/                  # Wazuh rules and settings
│   ├── thehive/                # TheHive templates and configs
│   ├── cortex/                 # Cortex analyzer configurations
│   └── shuffle/                # SOAR workflows and playbooks
├── scenarios/                   # Training scenario definitions
│   ├── basic/                  # Entry-level scenarios
│   ├── intermediate/           # Mid-level complexity
│   ├── advanced/               # Expert-level scenarios
│   └── false-positives/        # Benign activity scenarios
├── agents/                      # Simulated endpoint agents
│   ├── linux-agent/           # Linux host simulator
│   ├── windows-agent/         # Windows host with Sysmon
│   └── network-simulator/     # Network traffic generator
├── data/                        # Persistent data storage
│   ├── elasticsearch/         # ES data volume
│   ├── wazuh/                 # Wazuh data persistence
│   └── thehive/               # Case management data
└── scripts/                     # Utility scripts
    ├── backup.sh              # Data backup utilities
    ├── restore.sh             # Data restoration
    └── monitoring.sh          # Health check scripts
```

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Deployment Strategy

1. **Prerequisites Check**: Automated verification of Docker and system requirements
2. **Container Orchestration**: Single command deployment using Docker Compose
3. **Initial Configuration**: Automated setup of default configurations and integrations
4. **Scenario Loading**: Dynamic scenario deployment based on training requirements
5. **Health Monitoring**: Continuous monitoring of all services with automated recovery

Success Metrics

- **Deployment Time**: Complete setup in under 10 minutes on standard hardware
- **Resource Efficiency**: Operational on systems with 8GB RAM minimum
- **Scenario Variety**: Minimum 15 distinct training scenarios across difficulty levels

- **Documentation Quality**: Complete setup achievable by following documentation alone
- **Community Adoption**: Easy fork and customization for different training needs

This Virtual SOC platform will provide comprehensive, hands-on security training while maintaining simplicity in deployment and management, making advanced security operations training accessible to teams regardless of their infrastructure complexity.